

WILDENRATH, Germany

WMO: 104020

Lat: 51.117N

Lon: 6.217E

Elev: 292

StdP: 14.54

Time Zone: 1.00 (EUC)

Period: 82-91

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	13.7	18.5	6.7	7.8	15.3	11.9	10.2	21.0	29.5	47.9	26.7	45.8	6.4	60	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.2	84.1	67.1	80.9	65.7	77.5	64.3	69.0	80.5	67.3	77.8	65.7	75.0	7.1	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.1	93.9	73.1	63.4	88.4	71.3	62.0	83.9	69.9	33.2	80.9	31.9	78.2	30.6	75.2	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.8	20.1	17.7	DB	11.5	90.6	9.8	3.6	4.5	93.2	-1.2	95.3	-6.7	97.4	-13.8	100.0	(4)
(5)			WB	10.4	72.3	9.4	1.4	3.7	73.3	-1.8	74.2	-7.0	75.0	-13.8	76.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	49.8	35.8	35.3	42.4	47.4	55.1	59.6	64.5	64.0	58.2	52.4	42.7	39.3	(6)
(7)		DBStd	12.16	9.12	8.51	6.33	6.33	6.55	6.37	5.89	5.25	5.41	5.84	6.65	6.91	(7)
(8)		HDD50	1861	440	413	248	124	25	2	0	0	2	40	231	336	(8)
(9)		HDD65	5710	904	831	702	529	314	190	85	84	214	392	669	796	(9)
(10)		CDD50	1794	1	2	11	45	182	289	450	434	248	115	12	5	(10)
(11)		CDD65	165	0	0	0	0	6	26	69	53	10	1	0	0	(11)
(12)		CDH74	1546	0	0	0	2	74	240	652	483	89	6	0	0	(12)
(13)		CDH80	367	0	0	0	0	2	43	183	126	13	0	0	0	(13)
(14)	Wind	WSAvg	7.5	9.9	8.5	8.4	6.9	6.1	6.5	6.3	6.0	6.6	7.7	8.0	8.5	(14)
(15)	Precipitation	PrecAvg	30.0	2.5	2.3	2.0	1.7	2.3	2.8	3.0	3.1	2.3	2.4	2.6	2.9	(15)
(16)		PrecMax	38.3	5.4	5.4	3.7	4.1	4.4	7.4	6.4	6.4	6.3	6.1	4.3	6.2	(16)
(17)		PrecMin	23.5	0.2	0.4	0.1	0.0	0.7	0.6	1.3	0.7	0.5	1.3	0.2	1.0	(17)
(18)		PrecStd	3.7	1.2	1.2	1.0	0.9	1.0	1.5	1.4	1.7	1.5	0.9	1.0	1.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.6	59.2	65.9	72.7	79.7	84.2	89.2	82.3	73.8	60.8	57.2	(19)	
(20)			MCWB	49.8	50.2	53.3	57.3	61.0	67.1	69.0	66.7	65.5	63.0	54.4	51.7	(20)
(21)		2%	DB	51.0	54.0	59.2	67.7	76.2	80.8	84.8	83.4	76.3	68.1	56.9	54.3	(21)
(22)			MCWB	47.7	49.0	51.0	53.5	60.5	65.6	68.3	66.2	64.1	59.4	52.5	50.2	(22)
(23)		5%	DB	48.9	49.5	54.5	63.1	72.4	76.3	81.4	79.3	71.7	64.1	54.6	51.6	(23)
(24)			MCWB	46.4	46.0	48.7	52.2	58.2	63.5	66.7	65.3	62.0	57.5	51.2	48.5	(24)
(25)		10%	DB	47.1	46.6	51.5	59.1	68.1	71.8	77.3	75.4	67.9	61.1	52.3	49.0	(25)
(26)			MCWB	44.8	43.6	47.0	49.8	57.0	62.2	65.4	63.3	60.7	55.8	49.5	46.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.3	52.3	54.1	58.8	64.3	70.5	71.6	69.6	68.3	63.8	55.8	53.2	(27)
(28)			MCD B	53.0	57.3	63.0	71.1	75.9	79.1	84.5	82.7	77.5	71.8	58.6	55.1	(28)
(29)		2%	WB	48.6	49.2	52.0	55.4	61.5	67.4	69.6	67.6	65.6	60.4	53.5	50.6	(29)
(30)			MCD B	50.1	52.7	58.3	63.4	72.3	77.0	82.0	79.2	73.4	65.4	56.3	53.3	(30)
(31)		5%	WB	46.7	46.6	49.8	52.5	59.3	65.1	67.9	66.2	63.4	58.5	51.6	48.9	(31)
(32)			MCD B	48.5	49.3	53.1	61.1	69.8	73.6	78.9	77.1	69.4	63.0	54.0	51.3	(32)
(33)		10%	WB	44.7	44.0	47.9	50.7	57.7	62.6	66.0	64.5	61.6	56.6	49.9	47.0	(33)
(34)			MCD B	47.2	46.4	51.3	58.5	66.7	69.5	75.3	73.3	67.1	60.6	52.4	48.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.1	11.4	13.1	16.3	16.8	15.6	17.2	17.8	15.5	13.2	10.1	8.5	(35)
(36)			MCD BR	9.6	14.3	17.8	23.0	24.2	22.8	24.1	24.7	22.3	18.0	12.9	9.7	(36)
(37)			MCWBR	8.4	10.9	11.4	12.0	11.8	10.9	10.8	10.8	11.7	11.0	9.6	8.1	(37)
(38)		5% WB	MCD BR	9.3	13.7	15.6	21.1	21.3	21.1	21.4	22.0	19.6	15.9	11.7	9.2	(38)
(39)			MCWBR	8.8	11.2	10.8	11.7	11.4	11.2	10.2	10.4	11.5	10.3	9.3	8.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.323	0.346	0.377	0.412	0.418	0.418	0.424	0.415	0.387	0.361	0.337	0.319	(40)	
(41)		taud	2.410	2.370	2.289	2.214	2.237	2.254	2.236	2.285	2.353	2.419	2.408	2.380	(41)	
(42)		Ebn at Noon	221	246	258	262	266	266	262	258	252	237	212	201	(42)	
(43)		Edh at Noon	20	26	34	41	42	41	42	37	31	25	19	17	(43)	
(44)	All-Sky Solar Radiation	RadAvg	237	452	811	1300	1556	1689	1600	1366	1007	592	278	181	(44)	
(45)		RadStd	35	87	121	148	142	150	195	130	115	54	35	25	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (86 neighbors)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page